

PENGWIN 2026 Task 1 — V1 STU-Net Two-Stage Submission

Algorithm Description
2026-06-06

0. Submission intent

This is the **V1** submission for PENGWIN 2026 Task 1: per-fragment instance segmentation of pelvic bone fractures (sacrum, left/right hip, femur) in CT. It supersedes the V0.3.x pipeline-test (a ResEnc, pelvic-only, femur-stub container). V1 ships a ****STU-Net two-stage anatomy-conditioned pipeline with femur fully modeled****, trained on a leakage-free, case-grouped split.

1. Method overview

A two-stage, anatomy-conditioned cascade wrapped in a 4-layer robustness shell for Grand Challenge out-of-distribution defense.

- **Stage 1 — anatomy** (Dataset539_PelvicFemurAnatomyV3): a whole-CT 5-class semantic model (``0=bg, 1=sacrum, 2=leftHip, 3=rightHip, 4=femur``).
- **Per-anatomy ROI**: for each present bone, take the bbox of the Stage-1 probability ``>= 0.5`` (padded), and build a ****3-channel ROI**** — bone-LUT CT, Stage-1 anatomy probability, and a signed distance map clipped to ± 40 mm.
- **Stage 2 — fracture** (Dataset538_PelvicFemurBICMFractureV5): a per-anatomy ABBC model that emits a 4-class field (``background / border / boundary / core``).
- **Decoder**: core-seed watershed — connected components of the core class are watershed seeds; support is flooded from them; a ≥ 1 cm³ component prune and a small-fragment merge follow.
- **Assembly**: per-bone fragment IDs are offset into the official PENGWIN ranges (sacrum 1-50, leftHip 51-100, rightHip 101-150, ****femur 151-200****) and the volume is reoriented to the input frame.

Backbone. Both stages use **STU-Net-B** (58 M params, Apache-2.0) warm-started from a TotalSegmentator bone pretrain — a license-clean transfer that initializes the encoder for pelvic/femoral bone. (This replaces the former ResEnc-L.)

2. Stage 1 — Ds539 anatomy

- STU-Net-B, nnUNetResEncUNetLPlans / 3d_fullres, 1-channel CT input, 5-class softmax. Trainer ``PenguinTrainerSTUNetBaseAnatomyV301``.
- Input: LPS canonicalize + bone HU window + bone-LUT normalize.
- Output: full-CT 5-class anatomy probability, resampled back to the CT grid.

3. Pelvic-vs-femur routing

Pelvic cases (sacrum + both hips) and femur cases are disjoint patients. The case is routed **after** Stage-1 inference, by the femur-vs-pelvic argmax volume ratio of the Ds539 prediction (env-tunable threshold ~ 0.45). This replaces the spacing-only heuristic of V0.3.x (~ 50 % accurate, which would mis-route most femur cases now that femur is modeled).

4. Stage 2 — Ds538 ABBC fracture

- STU-Net-B, 3-channel input, 4-class ABBC head. Trainer ``...DenseCandidateCore025StrongPeakNoContactABBCSTUNetBV301`` (a V300 boundary-attention refinement on the STU-Net backbone). Aligned with the PENGWIN 2024 1st-place (MIC-DKFZ) ABBC formulation: explicit contact-voxel classification is dropped in favour of a core-seed $\rightarrow$ boundary watershed.

- Training: unified **case-grouped** 5-fold split (seed 12345) shared with Stage 1, so a held-out fold is held out across *both* stages (no patient leakage; the held-out anatomy context is out-of-fold). Early-stopping on the boundary fragment proxy.

5. Core-seed watershed decoder

Core class -> `scipy.ndimage.label` seeds; watershed flood over the support mask; $\geq 1 \text{ cm}^3$ connected-component prune; small-fragment merge.

6. 4-layer robustness

L1) bone-skeleton anatomy decomposition ($\text{HU} > 200 + 3\text{D CC}$) — distribution-independent fallback. L2) Ds539 argmax masks — mutually-exclusive anatomy assignment. L3) post-pad bbox sanity ($\leq 50\%$ of volume) — rejects OOD masks. L4) 480 s time budget — guarantees the 10-minute GC limit. Largest-CC-keep per anatomy.

7. Performance (held-out, dev proxy)

Held-out grouped fold-0 val ($n=132$ per-anatomy ROIs), scored with our PENGWIN-2026 **official-aligned v2 proxy** (per-anatomy argmax IoU ≥ 0.10 ; Fracture Dice / Instance-F1 / HD95 / ASSD):

metric	overall	Femur	RightHip	LeftHip	Sacrum
Fracture Dice	**0.799**	0.845	0.854	0.759	0.732
Instance F1	**0.919**	0.953	0.950	0.876	0.892
HD95 (mm)	**18.9**	7.4	25.7	25.7	18.3

The decoder applies an **anatomy-specific** over-segmentation control: the sacrum (a single dominant bone whose predicted core can speckle into spurious islands) uses an aggressive size-ratio + min-component merge, while the multi-fragment hips/femur keep the defaults — lifting Sacrum Instance-F1 from 0.585 to 0.892 and overall F1 from 0.844 to 0.919 with no change to the other bones.

The official Grand-Challenge evaluator is unpublished; this is an aligned proxy. End-to-end per-case runtime measured 45-205 s (RTX 3090), well within the T4 / 10-minute budget.

8. Hardware + runtime

NVIDIA T4 16 GiB; STU-Net-B inference $\sim 4 \text{ GiB/stage}$; per-case 45-205 s, capped at 480 s. Container: PyTorch 2.1.2 + CUDA 11.8 + nnUNetv2 2.5.1.

9. Limitations

- **Dev measurement only** — the official test set is unreleased; numbers are a held-out dev proxy (Stage-1 anatomy context is in-fold for training cases), not a leaderboard result.
- Sacrum recall trades down slightly (0.835) from the anatomy-specific merge; genuinely multi-fragment sacra are the harder remaining case.
- Single fold, single seed; no ensembling.

10. References

- nnU-Net v2: Isensee, F. et al. **Nat Methods** 18, 203-211 (2021).
- STU-Net (TotalSegmentator bone pretrain).
- PENGWIN 2026 Task 1 baseline: github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline
- PENGWIN 2024 1st place (ABBC reference): MIC-DKFZ.